Kaleb Coleman

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SUMMARY

M.S. AI Engineering (Robotics) student at Arizona State University researching bimanual manipulation in NVIDIA Isaac Sim and energy-aware quadrotor control in MuJoCo. Background in deep learning, GPU/HPC experimentation, data engineering, and cybersecurity; focused on reproducible simulation and autonomous systems.

EDUCATION

Arizona State University Expected May 2028
M.S., Artificial Intelligence Engineering (Robotics)

Northern Arizona University May 2026
B.S., Data Science; Cybersecurity minor

TECHNICAL SKILLS

Robotics / Control: NVIDIA Isaac Sim, MuJoCo, USD scene authoring, UR5e/RG2 manipulation, inverse kinematics, model predictive control, system identification, collision and contact validation

ML / AI: PyTorch, scikit-learn, XGBoost, PyGAM, CNNs, autoencoders, VAEs, diffusion models, PCA, clustering, regression, computer vision

Languages / Data: Python, SQL, R, C, JavaScript, TypeScript; pandas, NumPy, SQLite, PostgreSQL, API ingestion, schema validation, rerunnable ETL pipelines

Systems / Tools: Linux, Git/GitHub, Docker, GPU acceleration, Slurm/HPC workflows, Selenium, Streamlit

ROBOTICS RESEARCH

Two-Arm Manipulation in NVIDIA Isaac Sim | RISE Lab 2026 - Present

- Developed a simulation setup with opposing UR5e arms and RG2 grippers manipulating a shared bar; added collision validation, joint-frame corrections, object-space goals, and per-arm inverse kinematics.
- Validated a 26-second coordinated-motion run with an initially held 0.1 kg bar: four-finger contact at 100% of measured samples, zero unexpected collisions, 0.135 mm maximum slip, and 2.02 mm maximum position tracking error.
- Preserved scene assets, experimental evidence, failure cases, and reproducible setup instructions; physical digital-twin calibration remains in progress.

Energy-Aware Prescribed-Time Quadrotor Control 2026 - Present

- Collaborating with John Lok on an S500-based MuJoCo test bench studying tracking deadlines, motor lag, drag, wind estimation, and system identification.
- Built reproducible simulation comparisons, saved-output baselines, and solver timing diagnostics; contributed to a paper draft. Results are simulation-based; hardware flight validation remains future work.

SELECTED MACHINE LEARNING & DATA PROJECTS

Image Reconstruction and Generative Modeling Spring 2026

- Built PyTorch autoencoder, denoising, VAE, and diffusion experiments on MNIST-family datasets; compared nonlinear reconstruction with PCA baselines using cross-validation.
- Implemented DDPM-style UNet, timestep embeddings, noise scheduling, and reverse sampling; organized reproducible GPU/Slurm runs, isolated outputs, and parameter sweeps.

Cross-Sport Shot Analysis | NAU Capstone 2025 - 2026

- Engineered NBA/NHL ingestion, validation, relational storage, and idempotent upserts across approximately 3 million shot records from 2014-2024.
- Built generalized additive models and a Shot Difficulty Index; visualized shot quality and scoring efficiency and presented the research at NAU UGRADS.

NAU Course Catalog Scraper | IAAAI Collaboration Spring 2026

- Automated PDF prefix extraction and Selenium collection of course records; exported cleaned CSV datasets and supported AI/ethics curriculum discovery with classification and R Markdown reporting.

PROFESSIONAL EXPERIENCE

Information Systems Security Analyst Intern | Gila River Indian Community Jun 2026 - Present

- Supported enterprise endpoint and SaaS security administration, threat-telemetry monitoring, email/phishing defenses, and secure remote troubleshooting.

AWARDS

NAU ASA DataFest 2026 - Best Visualization Award: Collaborative analysis and visual storytelling in a 48-hour data science competition.